

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12383936
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Hasegawa; Tomoya et al.

Cleaning method and processing apparatus

Abstract

A cleaning method for removing a deposit in a processing chamber is provided. The cleaning method includes adjusting a temperature in the processing chamber to a first temperature; supplying a first gas including a hydrogen fluoride gas into the processing chamber in which the temperature is adjusted to the first temperature; adjusting the temperature in the processing chamber to a second temperature that is higher than the first temperature; and supplying a second gas including the hydrogen fluoride gas and an ammonia gas into the processing chamber in which the temperature is adjusted to the second temperature.

Inventors:	Hasegawa; Tomoya (Yamanashi, JP), Sasaki; Koji (Yamanashi, JP)
Applicant:	Tokyo Electron Limited (Tokyo, JP)
Family ID:	88977878
Assignee:	Tokyo Electron Limited (Tokyo, JP)
Appl. No.:	18/315722
Filed:	May 11, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230390810 A1	Dec. 07, 2023

Foreign Application Priority Data

JP	2022-091146	Jun. 03, 2022
----	-------------	---------------

Publication Classification

Int. Cl.: B08B5/00 (20060101); B08B7/00 (20060101)

U.S. Cl.:

CPC **B08B5/00** (20130101); **B08B7/0064** (20130101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2019/0067006	12/2018	Hawrylchak	N/A	H01J 37/32724

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2014-068045	12/2013	JP	N/A

OTHER PUBLICATIONS

Google Patents translation of JP2014068045A (Year: 2024). cited by examiner

Primary Examiner: Golightly; Eric W

Attorney, Agent or Firm: IPUSA, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based on and claims priority to Japanese Patent Application No. 2022-091146, filed on Jun. 3, 2022, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The disclosure herein relates a cleaning method and a processing apparatus.

2. Description of the Related Art

(3) A technique by which deposits adhering to the interior of a processing chamber due to a film deposition process and the like are removed is known (see Patent Document 1, for example). In Patent Document 1, silicofluoride generated when the interior of the processing chamber is cleaned is oxidized by an oxidizing gas, and the oxidized silicofluoride is removed by hydrogen fluoride.

RELATED-ART DOCUMENTS

Patent Documents

(4) Patent Document 1: Japanese Laid-open Patent Application Publication No. 2014-068045

SUMMARY OF THE INVENTION

(5) According to an embodiment of the present disclosure, a cleaning method for removing a deposit in a processing chamber is provided. The cleaning method includes adjusting a temperature in the processing chamber to a first temperature; supplying a first gas including a hydrogen fluoride gas into the processing chamber in which the temperature is adjusted to the first temperature; adjusting the temperature in the processing chamber to a second temperature that is higher than the

first temperature; and supplying a second gas including the hydrogen fluoride gas and an ammonia gas into the processing chamber in which the temperature is adjusted to the second temperature.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Other objects and further features of the present invention will be apparent from the following detailed description when read in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 is a schematic view of a processing apparatus according to an embodiment;
- (3) FIG. 2 is a flowchart illustrating a cleaning method according to an embodiment;
- (4) FIG. 3 is a timing chart illustrating the cleaning method according to the embodiment; and
- (5) FIG. 4 is a timing chart illustrating one cycle of a second cleaning process.

DESCRIPTION OF THE EMBODIMENTS

(6) In the following, non-limiting exemplary embodiments of the present disclosure will be described with reference to the accompanying drawings. In the accompanying drawings, the same or corresponding members or components are denoted by the same or corresponding reference numerals and the description thereof will not be repeated.

(7) [Processing Apparatus]

(8) A processing apparatus **1** according to an embodiment will be described with reference to FIG. 1. As illustrated in FIG. 1, the processing apparatus **1** is a batch-type apparatus that performs a process on a plurality of substrates **W** at the same time. The substrates **W** may be, for example, semiconductor wafers. The process may include a film deposition process. The process may include an etching process.

(9) The processing apparatus **1** includes a processing chamber **10**, a gas supply **30**, an exhaust device **40**, a heater **50**, and a controller **90**.

(10) The interior of the processing chamber **10** can be depressurized. The processing chamber **10** accommodates the substrates **W**. The processing chamber **10** includes an inner tube **11** and an outer tube **12**. The inner tube **11** has a cylindrical shape with an open top and an open bottom. The outer tube **12** has a cylindrical shape with an open bottom, has a ceiling on the top, and covers the outer periphery of the inner tube **11**. The inner tube **11** and the outer tube **12** are arranged to be concentric to form a double tube structure. The inner tube **11** and the outer tube **12** are both formed of a heat-resistant material such as quartz, for example.

(11) The lower end of the processing chamber **10** is airtightly supported by a manifold **13**. The manifold **13** has a cylindrical shape. The manifold **13** is formed of, for example, stainless steel. A flange **14** is formed on the upper end of the manifold **13**, and the lower end of the outer tube **12** is disposed on and supported by the flange **14**. A seal member **15**, such as an O-ring, is interposed between the flange **14** and the lower end of the outer tube **12** to cause the interior of the outer tube **12** to be airtight.

(12) A support **16** having an annular shape is provided on the inner wall of the manifold **13**, and the lower end of the inner tube **11** is disposed on and supported by the support **16**.

(13) An opening is formed in the lower end of the manifold **13**, and a lid **17** is airtightly attached to the lower end of the manifold **13** through a seal member **18**, such as an O-ring, such that an opening in the lower end of the processing chamber **10**, that is, the opening in the manifold **13** is sealed. The lid **17** is formed of, for example, stainless steel.

(14) A rotating shaft **20**, which rotatably supports a boat **19** through a ferrofluidic seal (not illustrated), is attached to the center of the lid **17** so as to pass through the lid **17**. The lower end of the rotating shaft **20** is rotatably supported on an arm **22** of a lifting mechanism **21** that includes a boat elevator.

(15) A rotating plate **23** is provided on the upper end of the rotating shaft **20**, and the boat **19** is

provided above the rotating plate **23** via a heated platform **24** made of quartz. With the configuration, the lid **17** and the boat **19** are integrally moved up and down by raising and lowering the arm **22** of the lifting mechanism **21**. Thus, the boat **19** can be inserted into and removed from the processing chamber **10**. The boat **19** can be accommodated in the processing chamber **10**. The boat **19** substantially horizontally holds the plurality of, for example, 50 to 150 substrates **W**, such that the substrates **W** are spaced apart from one another when viewed in the vertical direction.

(16) The gas supply **30** is configured to introduce various processing gases into the processing chamber **10**. The gas supply **30** includes a hydrogen fluoride supply **31**, an ammonia supply **32**, a nitrogen supply **33**, and a deposition gas supply (not illustrated).

(17) The hydrogen fluoride supply **31** includes a hydrogen fluoride supply pipe **31a** in the processing chamber **10**, and includes a hydrogen fluoride supply path **31b** outside the processing chamber **10**. A hydrogen fluoride source **31c**, a mass flow controller **31d**, and a hydrogen fluoride valve **31e** are sequentially provided on the hydrogen fluoride supply path **31b** in order from the upstream side to the downstream side in the gas flow direction. With this configuration, a supply timing of a hydrogen fluoride (HF) gas in the hydrogen fluoride source **31c** is controlled by the hydrogen fluoride valve **31e**, and further the flow rate of the hydrogen fluoride gas is adjusted to a predetermined flow rate by the mass flow controller **31d**. The hydrogen fluoride gas flows from the hydrogen fluoride supply path **31b** into the hydrogen fluoride supply pipe **31a**, and is discharged from the hydrogen fluoride supply pipe **31a** into the processing chamber **10**.

(18) The ammonia supply **32** includes an ammonia supply pipe **32a** in the processing chamber **10**, and includes an ammonia supply path **32b** outside the processing chamber **10**. An ammonia source **32c**, a mass flow controller **32d**, and an ammonia valve **32e** are sequentially provided on the ammonia supply path **32b** in order from the upstream side to the downstream side in the gas flow direction. With this configuration, a supply timing of an ammonia (NH₃) gas in the ammonia source **32c** is controlled by the ammonia valve **32e**, and further the flow rate of the ammonia gas is adjusted to a predetermined flow rate by the mass flow controller **32d**. The ammonia gas flows from the ammonia supply path **32b** into the ammonia supply pipe **32a**, and is discharged from the ammonia supply pipe **32a** into the processing chamber **10**.

(19) The nitrogen supply **33** includes a nitrogen supply pipe **33a** in the processing chamber **10**, and includes a nitrogen supply path **33b** outside the processing chamber **10**. A nitrogen source **33c**, a mass flow controller **33d**, and a nitrogen valve **33e** are sequentially provided on the nitrogen supply path **33b** in order from the upstream side to the downstream side in the gas flow direction. With this configuration, a supply timing of a nitrogen (N₂) gas in the nitrogen source **33c** is controlled by the nitrogen valve **33e**, and further the flow rate of the nitrogen gas is adjusted to a predetermined flow rate by the mass flow controller **33d**. The nitrogen gas flows from the nitrogen supply path **33b** into the nitrogen supply pipe **33a**, and is discharged from the nitrogen supply pipe **33a** into the processing chamber **10**. The nitrogen gas is an example of an inert gas.

(20) Each of the gas supply pipes (the hydrogen fluoride supply pipe **31a**, the ammonia supply pipe **32a**, and the nitrogen supply pipe **33a**) is formed of, for example, quartz. Each of the gas supply pipes is fixed to the manifold **13**. Each of the gas supply pipes extends linearly in the vicinity of the inner tube **11** along the vertical direction, bends in an L-shape in the manifold **13**, and extends in the horizontal direction so as to pass through the manifold **13**. The gas supply pipes are provided along the circumferential direction of the inner tube **11** at the same height. Each of the gas supply pipes has an opening in its tip located inside the inner tube **11**, and discharges the gas upward through the opening within the processing chamber **10**.

(21) A heater **31f** is attached to the hydrogen fluoride supply pipe **31a** and the hydrogen fluoride supply path **31b**. The heater **31f** heats hydrogen fluoride flowing in the hydrogen fluoride supply pipe **31a** and the hydrogen fluoride supply path **31b** so as to suppress corrosion of the hydrogen fluoride supply pipe **31a** and the hydrogen fluoride supply path **31b** by the hydrogen fluoride gas. The heater **31f** includes, for example, a pipe heater, a cartridge heater, or a combination thereof.

The heater **31f** may be attached to either the hydrogen fluoride supply pipe **31a** or the hydrogen fluoride supply path **31b**.

(22) The deposition gas supply may have a configuration similar to those of the other supplies (the hydrogen fluoride supply **31**, the ammonia supply **32**, and the nitrogen supply **33**). Similar to the other supplies, the deposition gas supply includes a deposition gas supply pipe, a deposition gas supply path, a deposition gas source, a mass flow controller, and a deposition gas valve.

(23) The gas supply **30** may mix different gases and discharge a gas mixture of the gases from one supply pipe. The gas supply pipes (the hydrogen fluoride supply pipe **31a**, the ammonia supply pipe **32a**, the nitrogen supply pipe **33a**, and the deposition gas supply pipe) may have different shapes and different arrangements. The gas supply may be configured to supply any other gas, in addition to the hydrogen fluoride gas, the ammonia gas, the nitrogen gas, and the deposition gas.

(24) The exhaust device **40** exhausts the gas that flows from the interior of the inner tube **11** through a space between the inner tube **11** and the outer tube **12** into a gas outlet **41**. The gas outlet **41** is formed in the side wall of the upper portion of the manifold **13** so as to be situated above the support **16**. An exhaust passage **42** is connected to the gas outlet **41**. A pressure regulating valve **43** and a vacuum pump **44** are sequentially provided on the exhaust passage **42** in order from the upstream side to the downstream side in the gas flow direction. The controller **90** controls the exhausting device **40** to operate the pressure regulating valve **43** and the vacuum pump **44**, and thus the pressure in the processing chamber **10** is controlled by the pressure regulating valve **43**, while the gas in the processing chamber **10** is suctioned by the vacuum pump **44**.

(25) The heater **50** includes a cylindrical heater **51** that surrounds the outer tube **12** and is located radially outward relative to the outer tube **12**. The heater **51** heats the substrates **W** accommodated in the processing chamber **10** by heating the entire outer periphery of the processing chamber **10**.

(26) The controller **90** may be implemented by a computer that includes one or more processors **91**, a memory **92**, an input/output interface (not illustrated), and an electronic circuit (not illustrated). Such a processor **91** may be a combination of one or more of a central processing unit (CPU), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a circuit with multiple discrete semiconductors, and the like. The memory **92** may include a volatile memory and a nonvolatile memory (for example, a compact disc, a digital versatile disc (DVD), a hard disk, a flash memory, and the like). The memory **92** stores a program that causes the processing apparatus **1** to operate, and stores a recipe such as processing conditions of substrate processing. By executing the program and the recipe stored in the memory **92**, the processor **91** controls each component of the processing apparatus **1** to perform a cleaning method as will be described below.

(27) [Cleaning Method]

(28) A cleaning method according to an embodiment will be described with reference to FIG. 2. As illustrated in FIG. 2, the cleaning method according to the embodiment includes a temperature lowering process **S10**, a first cleaning process **S20**, a first temperature raising process **S30**, a second cleaning process **S40**, and a second temperature raising process **S50**.

(29) In the temperature lowering process **S10**, the temperature in the processing chamber, maintained at a deposition temperature for performing the film deposition process, is lowered to a first temperature for performing the first cleaning process **S20**. The film deposition process may be a process for depositing, for example, a silicon oxide film. The deposition temperature is, for example, 350° C. The first temperature is, for example, a room temperature. The room temperature is a temperature in a clean room in which the processing chamber is placed, and is, for example, ° C.

(30) The first cleaning process **S20** is performed after the temperature lowering process **S10**. In the first cleaning process **S20**, a first gas that includes the hydrogen fluoride gas and does not include the ammonia gas is supplied into the processing chamber in which the temperature is adjusted to the first temperature. As a result, deposits that include silicon oxide adhering to the interior of the

processing chamber are removed. In the cleaning using the hydrogen fluoride gas, the etching rate is high. Therefore, deposits adhering to the interior of the processing chamber can be removed in a short period of time. However, in the cleaning using the hydrogen fluoride gas, as the temperature of the hydrogen fluoride gas increases, the etching rate decreases. Therefore, if an area that is higher in temperature than other areas is present in the processing chamber, some deposits may remain in the area without being etched. In the first cleaning process **S20**, the first gas may be heated to a third temperature and supplied. The third temperature is higher than the first temperature.

(31) The first temperature raising process **S30** is performed after the first cleaning process **S20**. In the first temperature raising process **S30**, the temperature in the processing chamber maintained at the first temperature is raised to a second temperature for performing the second cleaning process **S40**. The second temperature is higher than the first temperature, and is, for example, 65° C. or higher and 100° C. or lower.

(32) The second cleaning process **S40** is performed after the first temperature raising process **S30**. In the second cleaning process **S40**, deposits adhering to the interior of the processing chamber are removed by chemical oxide removal (COR) that performs chemical etching. Specifically, a second gas that includes the hydrogen fluoride gas and the ammonia gas is supplied into the processing chamber in which the temperature is adjusted to the second temperature. Then, the hydrogen fluoride gas and the ammonia gas are reacted with silicon oxide to produce ammonium silicofluoride ((NH₄)₂SiF₆), which is then sublimated by heating. Accordingly, deposits adhering to the interior of the processing chamber can be removed. In the COR, although the etching rate is low, the entirety of the interior of the processing chamber can be etched. Accordingly, deposits remaining without being etched in the first cleaning process **S20** can be removed.

(33) The second temperature raising process **S50** is performed after the second cleaning process **S40**. In the second temperature raising process **S50**, the temperature in the processing chamber maintained at the second temperature is raised to the deposition temperature for performing the film deposition process. Accordingly, the film deposition process can be started again in the processing chamber. The cleaning method according to the embodiment ends.

(34) In the cleaning method according to the above-described embodiment, the first cleaning process **S20** and the second cleaning process **S40** are performed in this order. In the first cleaning process **S20**, deposits adhering to the interior of the processing chamber are removed by the hydrogen fluoride gas. Therefore, deposits adhering to the interior of the processing chamber can be removed in a short period of time. However, in the first cleaning process **S20**, some deposits may remain without being etched in an area that is higher in temperature than other areas in the processing chamber. In the second cleaning process **S40**, deposits adhering to the interior of the processing chamber are removed by COR. Therefore, the entirety of the interior of the processing chamber can be etched, and thus deposits remaining without being etched in the first cleaning process **S20** can be removed. Accordingly, film residues remaining when removing deposits in the processing chamber can be reduced.

(35) Next, the operation of the processing apparatus **1** when the cleaning method is performed will be described with reference to FIGS. **3** and **4**. The cleaning method according to the embodiment will be performed after the film deposition process is performed one or more times by the processing apparatus **1**.

(36) First, the controller **90** controls the lifting mechanism **21** to raise the lid **17** and cause the lid **17** to contact the manifold **13**, such that the processing chamber **10** is airtightly sealed.

(37) Next, at a time point **t1**, the controller starts the temperature lowering process **S10**. At the time point **t1**, the controller **90** causes the heater **51** to stop heating. Accordingly, the temperature in the processing chamber **10**, maintained at the deposition temperature for performing the film deposition process, is gradually lowered and reaches the first temperature for performing the first

cleaning process S20. The deposition temperature is, for example, 350° C. The first temperature is a temperature in a clean room in which the processing apparatus 1 is placed, and, is for example, a room temperature. In the temperature lowering process S10, the controller 90 may use a coolant to forcibly cool the interior of the processing chamber 10.

(38) Next, at a time point t2 at which the temperature in the processing chamber 10 has reached the first temperature, the controller 90 causes the vacuum pump 44 to suction the gas in the processing chamber 10 while increasing the degree of opening of the pressure regulating valve 43 from 0%. As a result, the pressure in the processing chamber 10 gradually decreases from the atmospheric pressure (760 Torr), and reaches, for example, 30 Torr (4 kPa) or less.

(39) Next, at a time point t3 at which the pressure in the processing chamber 10 has reached 30 Torr or less, the controller 90 starts the first cleaning process S20. At the time point t3, the controller 90 opens the hydrogen fluoride valve 31e, and supplies the hydrogen fluoride gas from the hydrogen fluoride source 31c through the hydrogen fluoride supply path 31b and the hydrogen fluoride supply pipe 31a into the processing chamber 10. In addition, in accordance with the supply of the hydrogen fluoride gas, the controller 90 opens the nitrogen valve 33e, and supplies the nitrogen gas from the nitrogen source 33c through the nitrogen supply path 33b and the nitrogen supply pipe 33a into the processing chamber 10. By supplying the nitrogen gas in addition to the hydrogen fluoride gas, the uniformity of etching in the height direction of the processing chamber 10 can be easily controlled. Further, the controller 90 controls the mass flow controller 31d to adjust the flow rate of the hydrogen fluoride gas to, for example, 2 slm, and controls the mass flow controller 33d to adjust the flow rate of the nitrogen gas to, for example, 6 slm. At the time point t3, the controller 90 controls the degree of opening of the pressure regulating valve 43 and adjusts the pressure in the processing chamber 10 to, for example, 30 Torr (4 kPa). When the hydrogen fluoride gas is supplied from the hydrogen fluoride source 31c into the processing chamber 10, the controller 90 may operate the heater 31f to heat the hydrogen fluoride gas, flowing in the hydrogen fluoride supply pipe 31a and the hydrogen fluoride supply path 31b, to the third temperature. The third temperature may be a temperature that does not cause corrosion of the hydrogen fluoride supply pipe 31a and the hydrogen fluoride supply path 31b by hydrogen fluoride. Accordingly, corrosion of the hydrogen fluoride supply pipe 31a and the hydrogen fluoride supply path 31b by the hydrogen fluoride gas can be suppressed.

(40) Next, until a first period of time elapses from a time point t4 at which the pressure in the processing chamber 10 has reached 30 Torr, the controller 90 continues to supply the hydrogen fluoride gas and the nitrogen gas into the processing chamber 10. The first period of time may be determined in accordance with the amount of deposits adhering to the interior of the processing chamber 10.

(41) Next, at a time point t5 at which the first period of time has elapsed from the time point t4, the controller 90 ends the first cleaning process S20. At the time point t5, the controller 90 closes the hydrogen fluoride valve 31e and the nitrogen valve 33e, and stops the supply of the hydrogen fluoride gas and the nitrogen gas into the processing chamber 10. As a result, the pressure in the processing chamber 10 decreases.

(42) In the first cleaning process S20, deposits that include silicon oxide adhering to the interior of the processing chamber 10 are removed by the hydrogen fluoride gas being supplied into the processing chamber 10. In the cleaning using the hydrogen fluoride gas, the etching rate is high, and thus deposits can be removed in a short period of time. However, in the cleaning using the hydrogen fluoride gas, as the temperature of the hydrogen fluoride gas increases, the etching rate decreases. Therefore, if an area that is higher in temperature than other areas is present in the processing chamber 10, some deposits may remain in the area without being etched. For example, in a space between the inner tube 11 and the outer tube 12, an area, corresponding to the position where the hydrogen fluoride supply pipe 31a extends vertically, tends to be heated by the heat of the hydrogen fluoride gas heated by the heater 31f and flowing in the fluoride supply pipe 31a, and

thus, deposits tend to remain in the vicinity of the area.

(43) Next, at a time point **t6** at which the pressure in the processing chamber **10** has decreased, the controller **90** starts the first temperature raising process **S30**. At the time point **t6**, the controller **90** opens the nitrogen valve **33e**, and supplies the nitrogen gas from the nitrogen source **33c** through the nitrogen supply path **33b** and the nitrogen supply pipe **33a** into the processing chamber **10**. Further, the controller **90** controls the mass flow controller **33d** to adjust the flow rate of the nitrogen gas to, for example, 20 slm. As a result, the remaining hydrogen fluoride gas in the processing chamber **10** is replaced with the nitrogen gas.

(44) Next, at a time point **t7** at which a predetermined period of time elapses from the time point **t6**, the controller **90** operates the heater **51** to heat the interior of the processing chamber **10**. The temperature in the processing chamber **10** is gradually raised by the heating of the heater **51**, and reaches the second temperature for performing the second cleaning process **S40**. The second temperature is, for example, 100° C.

(45) Next, at a time point **t8** at which a predetermined period of time has elapsed from the time point **t7**, the controller **90** starts the second cleaning process **S40**. At the time point **t8**, while the controller **90** continues to supply the nitrogen gas into the processing chamber **10**, the controller **90** opens the hydrogen fluoride valve **31e**, and supplies the hydrogen fluoride gas from the hydrogen fluoride source **31c** through the hydrogen fluoride supply path **31b** and the hydrogen fluoride supply pipe **31a** into the processing chamber **10**. In addition, in accordance with the supply of the hydrogen fluoride gas, the controller **90** opens the ammonia valve **32e**, and supplies the ammonia gas from the ammonia source **32c** through the ammonia supply path **32b** and the ammonia supply pipe **32a** into the processing chamber **10**. Further, the controller **90** controls the mass flow controller **31d** to adjust the flow rate of the hydrogen fluoride gas to, for example, 0.9 slm. Further, the controller **90** controls the mass flow controller **32d** to adjust the flow rate of the ammonia gas to, for example, 3 slm. Further, the controller **90** controls the mass flow controller **33d** to adjust the flow rate of the nitrogen gas to, for example, 4 slm.

(46) Next, at a time point **t9** at which a predetermined period of time (for example, one minute) has elapsed from the time point **t8**, the controller **90** closes the hydrogen fluoride valve **31e** and the ammonia valve **32e**, and stops the supply of the hydrogen fluoride gas and the ammonia gas into the processing chamber **10**. The controller **90** controls the mass flow controller **33d** to adjust the flow rate of the nitrogen gas to, for example, 0.6 slm.

(47) Next, at a time point **t10** at which a predetermined period of time has elapsed from the time point **t9**, the controller **90** controls the mass flow controller **33d** to adjust the flow rate of the nitrogen gas to, for example, 6 slm. As a result, the remaining hydrogen fluoride gas and the remaining ammonia gas in the processing chamber **10** are replaced with the nitrogen gas.

(48) Next, at a time point **t11** at which a predetermined period of time has elapsed from the time point **t10**, the controller **90** closes the nitrogen valve **33e**, and stops the supply of the nitrogen gas into the processing chamber **10**.

(49) Next, at a time point **t12** at which a predetermined period of time has elapsed from the time point **t11**, the controller **90** opens the nitrogen valve **33e**, and supplies the nitrogen gas from the nitrogen source **33c** through the nitrogen supply path **33b** and the nitrogen supply pipe **33a** into the processing chamber **10**. Further, the controller **90** controls the mass flow controller **33d** to adjust the flow rate of the nitrogen gas to, for example, 6 slm. As a result, the remaining hydrogen fluoride gas and the remaining ammonia gas in the processing chamber **10** are replaced with the nitrogen gas.

(50) Next, at a time point **t13** at which a predetermined period of time has elapsed from the time point **t12**, the controller **90** closes the nitrogen valve **33e**, and stops the supply of the nitrogen gas into the processing chamber **10**. The controller **90** continues to stop the supply of the nitrogen gas into the processing chamber **10** until a time point **t14**.

(51) The controller **90** sets the time point **t8** to the time point **t14** as one cycle, repeats this cycle a

plurality of times, and ends the second cleaning process S40. As described, in the second cleaning process S40, the temperature in the processing chamber 10 is adjusted to the second temperature, and a step of simultaneously supplying the hydrogen fluoride and the ammonia gas into the processing chamber 10 and a step of supplying only the nitrogen gas into the processing chamber 10 are repeated a plurality of times.

(52) In the second cleaning process S40, deposits adhering to the interior of the processing chamber 10 are removed by COR. That is, the hydrogen fluoride gas and the ammonia gas are reacted with silicon oxide to produce ammonium silicofluoride ((NH₄)₂SiF₆), which is then sublimated by heating. Accordingly, deposits adhering to the interior of the processing chamber 10 can be removed. In the COR, although the etching rate is low, the entirety of the interior of the processing chamber 10 can be etched. Accordingly, deposits remaining without being etched in the first cleaning process S20 can be removed.

(53) Next, at a time point t14, the controller starts the second temperature raising process S50. At the time point t14, the controller 90 controls the mass flow controller 33d to adjust the flow rate of the nitrogen gas to, for example, 20 slm.

(54) Next, at a time point t15 at which a predetermined period of time has elapsed from the time point t14, the controller 90 operates the heater 51 to heat the interior of the processing chamber 10. The temperature in the processing chamber 10 is gradually raised by the heating of the heater 51, and reaches the deposition temperature for performing the film deposition process.

(55) Next, at a time point t16 at which the temperature in the processing chamber 10 has reached the deposition temperature, the controller 90 stops the suction of the gas in the processing chamber 10 by the vacuum pump 44. As a result, the pressure in the processing chamber 10 gradually increases and reaches the atmospheric pressure (760 Torr).

(56) Next, the controller 90 controls the lifting mechanism 21 to lower the lid 17 and cause the lid 17 to be spaced apart from the manifold 13, thereby opening the processing chamber 10.

(57) As described, in the cleaning method performed by the processing apparatus 1 according to the embodiment, deposits that include silicon oxide adhering to the interior of the processing chamber can be removed.

(58) In the cleaning method according to the above-described embodiment, the first cleaning process S20 and the second cleaning process S40 are performed in a state in which the boat 19 is not accommodated in the processing chamber 10; however, the cleaning method according to the above-described embodiment is not limited thereto. For example, the first cleaning process S20 and the second cleaning process S40 may be performed in a state in which the boat 19 is accommodated in the processing chamber 10. In this case, when deposits adhering to the interior of the processing chamber 10 are removed, deposits adhering to the boat 19 can also be removed.

(59) According to an embodiment of the present disclosure, film residues remaining when removing deposits in a processing chamber can be reduced.

(60) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosures. Indeed, the embodiments described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the disclosures. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosures.

Claims

1. A cleaning method for removing a deposit including silicon oxide in a processing chamber, the cleaning method comprising: adjusting a temperature in the processing chamber to a first temperature; supplying a first gas including a hydrogen fluoride gas that is heated to a third

temperature, into the processing chamber in which the temperature is adjusted to the first temperature, such that the deposit is removed from the processing chamber when reaching the first temperature, wherein the third temperature is higher than the first temperature; adjusting the temperature in the processing chamber to a second temperature that is higher than the first temperature; supplying a second gas including the hydrogen fluoride gas and an ammonia gas into the processing chamber in which the temperature is adjusted to the second temperature; reacting the hydrogen fluoride gas and the ammonia gas with the silicon oxide to produce ammonium silicofluoride; and sublimating the ammonium silicofluoride by heating, such that a portion of the deposit that remains unetched after supplying the first gas is removed from the processing chamber when reaching the second temperature.

2. The cleaning method according to claim 1, wherein the first temperature is a room temperature, and the second temperature is 65° C. or higher and 100° C. or lower.

3. The cleaning method according to claim 1, wherein the supplying of the second gas includes repeating a cycle including: simultaneously supplying the hydrogen fluoride gas and the ammonia gas into the processing chamber, and supplying an inert gas into the processing chamber without supplying the hydrogen fluoride gas and the ammonia gas.

4. A processing apparatus comprising: a processing chamber; a gas supply configured to supply a processing gas into the processing chamber; a heater configured to heat the processing chamber; and a controller configured to control the gas supply and the heater to perform a process including: adjusting a temperature in the processing chamber to a first temperature; supplying a first gas including a hydrogen fluoride gas that is heated to a third temperature, into the processing chamber in which the temperature is adjusted to the first temperature, such that a deposit is removed from the processing chamber when reaching the first temperature, wherein the third temperature is higher than the first temperature and the deposit includes silicon oxide; raising the temperature in the processing chamber to a second temperature that is higher than the first temperature; supplying a second gas including the hydrogen fluoride gas and an ammonia gas into the processing chamber in which the temperature is adjusted to the second temperature; reacting the hydrogen fluoride gas and the ammonia gas with the silicon oxide to produce ammonium silicofluoride; and sublimating the ammonium silicofluoride by heating, such that a portion of the deposit that remains unetched after supplying the first gas is removed from the processing chamber when reaching the second temperature.
